Lynbrook Math Summer Camp 2026
G2 Special Triangles

Lynbrook Math Club

August 2026

G2.1 Lecture

Triangles are everywhere in the real world and in mathematics. While calculating side lengths and angles can often involve annoying applications of the Pythagorean Theorem or trigonometry, certain triangles possess interesting properties that make them easy to work with. In this lesson, we will cover special triangles and clever ways to use their unique side length ratios to solve geometry problems quickly and elegantly.

Pythagorean Triples

Before diving into angle-based special triangles, let's revisit right triangles where all three side lengths are integers.

Definition G2.1.1 (Pythagorean Triple)**.** A **Pythagorean triple** is a tuple of three positive integers (a, b, c) such that $a^2 + b^2 = c^2$. Here, a and b represent the legs of a right triangle, and c represents the hypotenuse.

If (a, b, c) is a Pythagorean triple, then any integer scalar multiple (ka, kb, kc) is also a Pythagorean triple, forming a right triangle with proportional side lengths.

Claim G2.1.2 (Common Pythagorean Triples) — The following side length ratios appear frequently and are extremely useful to memorize:

- $3 : 4 : 5$ and its multiples like $(6 : 8 : 10), (9 : 12 : 15), \ldots$
- $5 : 12 : 13$
- $8 : 15 : 17$
- $7 : 24 : 25$

Example 3

A right triangle has a leg of length 18 and a hypotenuse of length 30. What is its area?

Instead of using the Pythagorean Theorem directly with large numbers ($30^2 - 18^2$), we look for a common ratio. Notice that $\gcd(18, 30) = 6$:

$$\frac{18}{6} = 3, \quad \frac{30}{6} = 5.$$

This is a scaled-up $3 : 4 : 5$ right triangle! The missing leg must be $\sqrt{5^2 - 3^2} \times 6 = 4 \times 6 = 24$.

Now, we can compute the area:

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 18 \times 24 = 216.$$

Isosceles Right Triangles ($45^\circ - 45^\circ - 90^\circ$)

When a right triangle is also isosceles, its two acute angles must sum to 90° and be equal, making each acute angle equal to 45°.

Definition G2.1.4 ($45^\circ - 45^\circ - 90^\circ$ Triangle)**.** A $45^\circ - 45^\circ - 90^\circ$ triangle is an isosceles right triangle. If the legs have length x, then by the Pythagorean Theorem, the hypotenuse has length:

$$\text{Hypotenuse} = \sqrt{x^2 + x^2} = \sqrt{2x^2} = x\sqrt{2}.$$

> **Claim G2.1.5** ($45^\circ - 45^\circ - 90^\circ$ Side Ratio) — The ratio of the side lengths opposite to the angles $45^\circ : 45^\circ : 90^\circ$ is always:
> $$1 : 1 : \sqrt{2}.$$

This means:

- To go from a **leg to the hypotenuse**, multiply by $\sqrt{2}$.
- To go from the **hypotenuse to a leg**, divide by $\sqrt{2}$ (or multiply by $\frac{\sqrt{2}}{2}$).

> **Example 6**
> A square has a diagonal of length 12. What is the area of the square?

Drawing the diagonal of a square splits it into two $45^\circ - 45^\circ - 90^\circ$ right triangles, where the diagonal acts as the hypotenuse and the sides of the square act as the legs.

Let s be the side length of the square. Then:

$$s\sqrt{2} = 12 \implies s = \frac{12}{\sqrt{2}} = \frac{12\sqrt{2}}{2} = 6\sqrt{2}.$$

The area of the square is simply s^2:

$$\text{Area} = (6\sqrt{2})^2 = 36 \times 2 = 72.$$

$30^\circ - 60^\circ - 90^\circ$ Triangles

Another common special right triangle comes directly from cutting an equilateral triangle in half along one of its altitudes.

Definition G2.1.7 ($30^\circ - 60^\circ - 90^\circ$ Triangle)**.** A $30^\circ - 60^\circ - 90^\circ$ triangle is a right triangle with acute angles measuring 30° and 60°.

If we start with an equilateral triangle of side length $2x$ and drop an altitude to split it in half:

- The side opposite the 30° angle is half the base, so its length is x.

- The hypotenuse (opposite the 90° angle) is the original side length, which is $2x$.
- The side opposite the 60° angle is the altitude. By the Pythagorean Theorem:

$$\text{Altitude} = \sqrt{(2x)^2 - x^2} = \sqrt{3x^2} = x\sqrt{3}.$$

Claim G2.1.8 ($30^\circ - 60^\circ - 90^\circ$ Side Ratio) — The ratio of the side lengths opposite to the angles $30^\circ : 60^\circ : 90^\circ$ is always:

$$1 : \sqrt{3} : 2.$$

Claim G2.1.9 (Equilateral Triangle Properties) — For an equilateral triangle with side length s:

- Height (Altitude) $h = \frac{s\sqrt{3}}{2}$
- Area $A = \frac{s^2\sqrt{3}}{4}$

Example 10

An equilateral triangle has a height of 9 cm. Find its perimeter and area.

The height splits the equilateral triangle into two $30^\circ - 60^\circ - 90^\circ$ triangles. The height is the leg opposite the 60° angle, so:

$$x\sqrt{3} = 9 \implies x = \frac{9}{\sqrt{3}} = 3\sqrt{3}.$$

Since x is the short leg (half the base), the full side length of the equilateral triangle is $s = 2x = 6\sqrt{3}$.

Therefore:

- **Perimeter:** $3 \times s = 3(6\sqrt{3}) = 18\sqrt{3}$ cm.
- **Area:** $\frac{s^2\sqrt{3}}{4} = \frac{(6\sqrt{3})^2\sqrt{3}}{4} = \frac{108\sqrt{3}}{4} = 27\sqrt{3}\text{ cm}^2$.

Combining Special Triangles

Geometry problems frequently stitch special right triangles together or hide them inside larger non-right triangles. The key strategy is often to **drop an altitude** to break down a complex shape into manageable $30^\circ - 60^\circ - 90^\circ$ or $45^\circ - 45^\circ - 90^\circ$ pieces.

Example 11

In $\triangle ABC$, $AB = 4\sqrt{2}$, $\angle A = 45^\circ$, and $\angle B = 30^\circ$. Find the exact length of side BC and the area of $\triangle ABC$.

Notice that $\triangle ABC$ is not a right triangle because $\angle C = 180^\circ - 45^\circ - 30^\circ = 105^\circ$. However, if we drop an altitude CD from vertex C perpendicular to side AB, we create two right triangles: $\triangle ADC$ and $\triangle BDC$.

1. **In $\triangle ADC$:** Since $\angle A = 45^\circ$ and $\angle ADC = 90^\circ$, $\triangle ADC$ is a $45^\circ - 45^\circ - 90^\circ$ triangle with height $h = CD$. Thus, $AD = h$.
2. **In $\triangle BDC$:** Since $\angle B = 30^\circ$, $\triangle BDC$ is a $30^\circ - 60^\circ - 90^\circ$ triangle. The long leg is $BD = h\sqrt{3}$, and the hypotenuse is $BC = 2h$.
3. **Finding h:** The total length of base AB is $AD + BD$:

$$h + h\sqrt{3} = 4\sqrt{2} \implies h(1+\sqrt{3}) = 4\sqrt{2}.$$

Solving for h:

$$h = \frac{4\sqrt{2}}{\sqrt{3}+1} = \frac{4\sqrt{2}(\sqrt{3}-1)}{2} = 2\sqrt{6} - 2\sqrt{2}.$$

4. **Finding BC and Area:**
   - $BC = 2h = 4\sqrt{6} - 4\sqrt{2}$.
   - Area $= \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 4\sqrt{2} \times (2\sqrt{6} - 2\sqrt{2}) = 8\sqrt{3} - 8$.

Example 12

A regular hexagon has a side length of 4. What is the area of the hexagon?

A regular hexagon can be partitioned into 6 congruent equilateral triangles sharing a central vertex.

Since the side length of the hexagon is 4, each equilateral triangle has side length $s = 4$. Using our equilateral triangle area formula:

$$\text{Area of 1 triangle} = \frac{s^2\sqrt{3}}{4} = \frac{4^2\sqrt{3}}{4} = 4\sqrt{3}.$$

Multiplying by 6 gives the total area of the hexagon:

$$\text{Total Area} = 6 \times 4\sqrt{3} = 24\sqrt{3}.$$